

```

import pandas as pd
import matplotlib.pyplot as mp
import seaborn as sb
import os

os.chdir(r'C:\Users\cebal\OneDrive\Documentos\Grandes Bases de Datos I')

from sklearn.cluster import KMeans

dataset = pd.read_csv('iris.csv')
dataset.head()

   sepal.length  sepal.width  petal.length  petal.width  variety
0           5.1           3.5           1.4           0.2   Setosa
1           4.9           3.0           1.4           0.2   Setosa
2           4.7           3.2           1.3           0.2   Setosa
3           4.6           3.1           1.5           0.2   Setosa
4           5.0           3.6           1.4           0.2   Setosa

# seleccionamos las variables que son numericas
fields = dataset.iloc[:, [0,1,2,3]].values
print(fields)

[[5.1 3.5 1.4 0.2]
 [4.9 3.  1.4 0.2]
 [4.7 3.2 1.3 0.2]
 [4.6 3.1 1.5 0.2]
 [5.  3.6 1.4 0.2]
 [5.4 3.9 1.7 0.4]
 [4.6 3.4 1.4 0.3]
 [5.  3.4 1.5 0.2]
 [4.4 2.9 1.4 0.2]
 [4.9 3.1 1.5 0.1]
 [5.4 3.7 1.5 0.2]
 [4.8 3.4 1.6 0.2]
 [4.8 3.  1.4 0.1]
 [4.3 3.  1.1 0.1]
 [5.8 4.  1.2 0.2]
 [5.7 4.4 1.5 0.4]
 [5.4 3.9 1.3 0.4]
 [5.1 3.5 1.4 0.3]
 [5.7 3.8 1.7 0.3]
 [5.1 3.8 1.5 0.3]
 [5.4 3.4 1.7 0.2]
 [5.1 3.7 1.5 0.4]]

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[4.6 3.6 1. 0.2]
[5.1 3.3 1.7 0.5]
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[6.2 2.9 4.3 1.3]
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[6.8 3.  5.5 2.1]
[5.7 2.5 5.  2. ]
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[7.2 3.2 6.  1.8]
[6.2 2.8 4.8 1.8]
[6.1 3.  4.9 1.8]
[6.4 2.8 5.6 2.1]
[7.2 3.  5.8 1.6]
[7.4 2.8 6.1 1.9]
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[6.4 2.8 5.6 2.2]
[6.3 2.8 5.1 1.5]
[6.1 2.6 5.6 1.4]
[7.7 3.  6.1 2.3]
[6.3 3.4 5.6 2.4]
[6.4 3.1 5.5 1.8]
[6.  3.  4.8 1.8]
[6.9 3.1 5.4 2.1]
[6.7 3.1 5.6 2.4]
[6.9 3.1 5.1 2.3]
[5.8 2.7 5.1 1.9]
[6.8 3.2 5.9 2.3]
[6.7 3.3 5.7 2.5]
[6.7 3.  5.2 2.3]
[6.3 2.5 5.  1.9]
[6.5 3.  5.2 2. ]
[6.2 3.4 5.4 2.3]
[5.9 3.  5.1 1.8]]
```

```
import warnings
warnings.filterwarnings('ignore')

wcss = []
for index in range(1,11):
    kmeans = KMeans(n_clusters = index, init = "k-means++", random_state = 2)
```


<matplotlib.legend.Legend at 0x1b7e85dd550><Figure size 1000x1000 with 1 Axes>

```
# Criterio de Silueta ("Silhouette")
from sklearn import datasets
from sklearn.metrics import silhouette_score

# Seleccionamos las cuatro variables num'ericas de la base original.
X = dataset.iloc[:, [0,1,2,3]].to_numpy()
for j in range(2,11):
    kmeans = KMeans(n_clusters = j, random_state = 42)
    kmeans.fit_predict(X)
    score = silhouette_score(X, kmeans.labels_, metric = 'euclidean')
    print("Score Silhouette:", "k =", j, ",", score)
```

```
Score Silhouette: k = 2 , 0.6810461692117465
Score Silhouette: k = 3 , 0.5511916046195927
Score Silhouette: k = 4 , 0.49764331793219296
Score Silhouette: k = 5 , 0.4930804067193529
Score Silhouette: k = 6 , 0.36784649847122536
Score Silhouette: k = 7 , 0.3542978877198859
Score Silhouette: k = 8 , 0.34467972180562056
Score Silhouette: k = 9 , 0.31558878533897866
Score Silhouette: k = 10 , 0.3014143745325154
```

```
# en este caso el Silhoutte con un valor mayor es el de k = 2
```

```
fields2 = dataset.iloc[:, [0,1,2,3]].values
print(fields2)
```

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[[5.1 3.5 1.4 0.2]
 [4.9 3.  1.4 0.2]
 [4.7 3.2 1.3 0.2]
 [4.6 3.1 1.5 0.2]
 [5.  3.6 1.4 0.2]
 [5.4 3.9 1.7 0.4]
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 [5.  3.4 1.5 0.2]
 [4.4 2.9 1.4 0.2]
 [4.9 3.1 1.5 0.1]
 [5.4 3.7 1.5 0.2]
 [4.8 3.4 1.6 0.2]
 [4.8 3.  1.4 0.1]
 [4.3 3.  1.1 0.1]
 [5.8 4.  1.2 0.2]
 [5.7 4.4 1.5 0.4]]
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[5.4 3.9 1.3 0.4]
[5.1 3.5 1.4 0.3]
[5.7 3.8 1.7 0.3]
[5.1 3.8 1.5 0.3]
[5.4 3.4 1.7 0.2]
[5.1 3.7 1.5 0.4]
[4.6 3.6 1. 0.2]
[5.1 3.3 1.7 0.5]
[4.8 3.4 1.9 0.2]
[5. 3. 1.6 0.2]
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[5.2 3.5 1.5 0.2]
[5.2 3.4 1.4 0.2]
[4.7 3.2 1.6 0.2]
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[5.3 3.7 1.5 0.2]
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[5.7 2.8 4.5 1.3]
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[4.9 2.4 3.3 1.]
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[5.9 3. 4.2 1.5]
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[5. 2.3 3.3 1.]
[5.6 2.7 4.2 1.3]
[5.7 3. 4.2 1.2]
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[6.2 2.9 4.3 1.3]
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[7.6 3. 6.6 2.1]

[4.9 2.5 4.5 1.7]
 [7.3 2.9 6.3 1.8]
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 [6.5 3.2 5.1 2.]
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 [5.8 2.8 5.1 2.4]
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 [6.5 3. 5.5 1.8]
 [7.7 3.8 6.7 2.2]
 [7.7 2.6 6.9 2.3]
 [6. 2.2 5. 1.5]
 [6.9 3.2 5.7 2.3]
 [5.6 2.8 4.9 2.]
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 [6.3 2.7 4.9 1.8]
 [6.7 3.3 5.7 2.1]
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 [6.2 2.8 4.8 1.8]
 [6.1 3. 4.9 1.8]
 [6.4 2.8 5.6 2.1]
 [7.2 3. 5.8 1.6]
 [7.4 2.8 6.1 1.9]
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 [6.4 2.8 5.6 2.2]
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 [6.7 3.3 5.7 2.5]
 [6.7 3. 5.2 2.3]
 [6.3 2.5 5. 1.9]
 [6.5 3. 5.2 2.]
 [6.2 3.4 5.4 2.3]
 [5.9 3. 5.1 1.8]]

```
# Inciso 3
fields2 = dataset.iloc[:, [0,1,2,3]].values
print(fields2)
```

```
[[5.1 3.5 1.4 0.2]
 [4.9 3.  1.4 0.2]
 [4.7 3.2 1.3 0.2]
 [4.6 3.1 1.5 0.2]
 [5.  3.6 1.4 0.2]
 [5.4 3.9 1.7 0.4]
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 [5.4 3.7 1.5 0.2]
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 [4.8 3.4 1.9 0.2]
 [5.  3.  1.6 0.2]
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[4.5 2.3 1.3 0.3]
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[6.5 3.2 5.1 2.]
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[6.8 3. 5.5 2.1]
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[5.8 2.8 5.1 2.4]
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[7.7 3.8 6.7 2.2]
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[6. 2.2 5. 1.5]
[6.9 3.2 5.7 2.3]
[5.6 2.8 4.9 2.]
[7.7 2.8 6.7 2.]
[6.3 2.7 4.9 1.8]
[6.7 3.3 5.7 2.1]
[7.2 3.2 6. 1.8]
[6.2 2.8 4.8 1.8]
[6.1 3. 4.9 1.8]
[6.4 2.8 5.6 2.1]
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[7.4 2.8 6.1 1.9]

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[7.9 3.8 6.4 2. ]
[6.4 2.8 5.6 2.2]
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[7.7 3.  6.1 2.3]
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[6.3 2.5 5.  1.9]
[6.5 3.  5.2 2. ]
[6.2 3.4 5.4 2.3]
[5.9 3.  5.1 1.8]]

```

```

from sklearn import decomposition
pca = decomposition.PCA(n_components = 2)
pca.fit(fields2)

```

```

fields2 = pca.transform(fields2)
fields2

```

```

array([[ -2.68412563,  0.31939725],
       [ -2.71414169, -0.17700123],
       [ -2.88899057, -0.14494943],
       [ -2.74534286, -0.31829898],
       [ -2.72871654,  0.32675451],
       [ -2.28085963,  0.74133045],
       [ -2.82053775, -0.08946138],
       [ -2.62614497,  0.16338496],
       [ -2.88638273, -0.57831175],
       [ -2.6727558 , -0.11377425],
       [ -2.50694709,  0.6450689 ],
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```

```

wcss2 = []
for index in range(1,11):

```

```

kmeans = KMeans(n_clusters = index,
                 init = "k-means++",
                 random_state = 2)
kmeans.fit(fields2)
wcsc2.append(kmeans.inertia_)

# Graficamos
sb.set()
mp.plot(range(1,11), wcsc2)
mp.title("K-Means Clustering despu'es de PCA")
mp.xlabel("N'umero de clusters")
mp.ylabel("WCSS")

mp.show()

<Figure size 640x480 with 1 Axes>

# Criterio de Silueta ("Silhouette") despu'es de PCA
for j in range(2,11):
    kmeans = KMeans(n_clusters = j, random_state = 42)
    kmeans.fit_predict(fields2)
    score = silhouette_score(fields2, kmeans.labels_, metric = 'euclidean')
    print("Score Silhouette:", "k =", j, ",", score)

Score Silhouette: k = 2 , 0.7056703225102109
Score Silhouette: k = 3 , 0.5976764219547994
Score Silhouette: k = 4 , 0.5577409232246783
Score Silhouette: k = 5 , 0.5100407194899199
Score Silhouette: k = 6 , 0.40319738737442673
Score Silhouette: k = 7 , 0.3868263997713608
Score Silhouette: k = 8 , 0.44076033249654045
Score Silhouette: k = 9 , 0.4020143876366366
Score Silhouette: k = 10 , 0.41588658017532376

# el numero de clusters recomendado por Silhouette se sigue manteniendo en 2 antes
# y despues
# de aplicar PCA, ademas el valor del indice Silhouette aumenta de .6810 a .7057 lo
# cual nos
# indica una mejor separacion de los grupos bajo este criterio.

```